
AWS Secrets Manager Overview

AWS Secrets Manager is a fully managed service that enables you to **securely store, retrieve, manage, and rotate secrets** such as:

- Database credentials
- API keys
- Tokens
- SSH keys
- Passwords

It helps eliminate hardcoding secrets in code, CI/CD pipelines, or config files.

Key Components of AWS Secrets Manager

Component	Description
 Secret	A key-value pair or JSON document that stores sensitive data like credentials.
 Automatic Rotation	Built-in support for rotation of secrets using AWS Lambda.
 Encryption	Secrets are encrypted at rest using AWS KMS (Key Management Service) .
 Policies & Permissions	Fine-grained access control using IAM policies and resource policies .
 Secret Versions	Supports versioning for secrets, which helps with rollback and audit.



Audit Logs

Integration with **AWS CloudTrail** for monitoring and auditing secret access.



Services Integrated with AWS Secrets Manager

Secrets Manager integrates seamlessly with many AWS services for both **retrieving secrets** and **auto-rotation**:



Database Integration

Supports built-in **rotation and retrieval** for these databases:

Database	Managed Rotation
Amazon RDS (MySQL, PostgreSQL, MariaDB)	✓
Amazon RDS (SQL Server, Oracle)	✓
Amazon Aurora	✓
Amazon Redshift	✓
Custom DBs	✓ (via Lambda)



Compute Services

To inject secrets dynamically at runtime:

- **Amazon EC2**  – Via instance profile or SDK to fetch secrets.
- **AWS Lambda**  – Pull secrets at runtime using AWS SDK.
- **Amazon ECS / EKS / Fargate**  – Use secrets in container environments via task definitions or Kubernetes secrets.



Developer Tools

- **AWS CodePipeline / CodeBuild**  – Fetch secrets during build/deploy.
- **AWS CloudFormation / CDK / Terraform** – Integrate secrets into templates without hardcoding.

Application Integration

Use AWS SDKs or Secrets Manager APIs in:

- Java, Python, Node.js, Go, .NET, etc.
- Retrieve secrets dynamically instead of using static credentials.

Security & Access Control

- **AWS IAM** – Enforce who can access/rotate/manage secrets.
 - **AWS KMS** – Encrypt/decrypt secrets.
 - **AWS CloudTrail** – Track usage and access logs.
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Secret Rotation Flow (with Lambda)

1.  Secrets Manager invokes a Lambda function.
2.  Lambda generates a new password.
3.  Lambda updates the DB or service with the new credentials.
4.  Lambda stores the new secret version in Secrets Manager.
5.  Rotation is seamless for the app if using dynamic retrieval.

Common Use Cases

Use Case	Example
 DB Credential Management	Rotate RDS secrets automatically every 30 days.
 Dev/Test Environments	Store different API keys per environment securely.
 CI/CD Pipelines	Use secure secrets in deployment jobs.
 Microservices	Centralized secret store for accessing third-party APIs.

Secrets Manager vs Systems Manager Parameter Store

Feature	Secrets Manager	SSM Parameter Store
Designed for secrets	✓	✗ (more for config)
Automatic rotation	✓	✗
Built-in encryption	✓ (KMS)	✓
Versioning	✓	✓
Cost	Higher	Lower (free tier available)

Example JSON Structure of a Secret

```
{  
  "username": "admin",  
  "password": "MySecureP@ss123"  
}
```

Best Practices

1.  **Enable rotation** for sensitive secrets.
 2.  **Use IAM policies** to restrict access.
 3.  **Audit access** with CloudTrail.
 4.  **Rotate secrets regularly** even for internal use.
 5.  **Never hardcode** secrets in code or config files.
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